Assignment 7: Hypothesis Testing using Language Models

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Codebase

The code, along with the data for this assignment is uploaded on GitHub: https://github.com/garguy07/hypothesis-testing-lm

Google Colab was used for running the code for this assignment. The code can be found at:

https://colab.research.google.com/drive/1bqOUt6n_KzrhNc9HDwjhL4EsoZKoWO-C?usp=sharing

Question 1: Computing Language Model Scores [10 Marks]

For each sentence in the BNC dataset, three scores were computed per model:

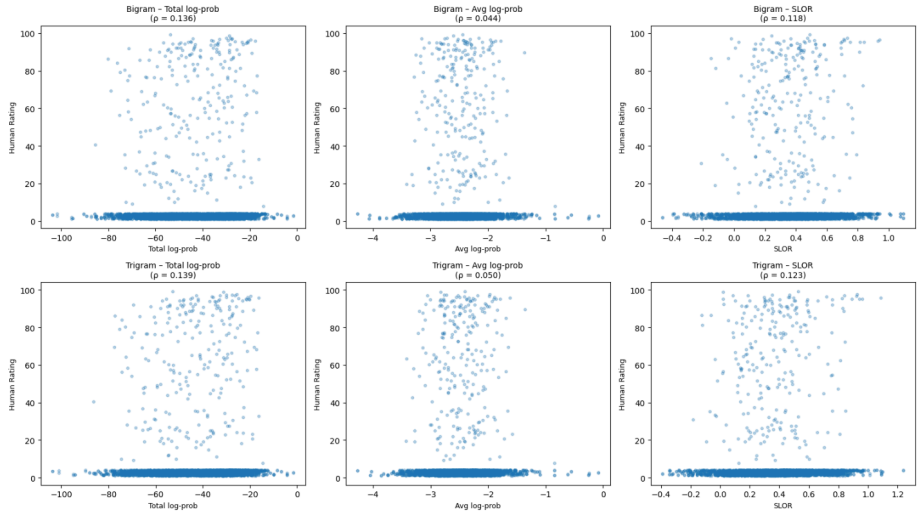
1. Total log-probability
2. Average per-word log-probability
3. SLOR (Syntactic Log-Odds Ratio)

Sample Scores (First 5 Sentences)

Sentence (truncated)	Human Rating	Bigram Total	Bigram Avg	Bigram SLOR	Trigram Total	Trigram Avg	Trigram SLOR
We've still got the Dower...	4.00	−34.56	−2.30	0.026	−34.43	−2.30	0.034
My first real friend was...	4.00	−33.31	−3.03	0.273	−33.03	−3.00	0.299
She was wearing denim...	4.00	−27.77	−2.31	0.711	−26.22	−2.19	0.840
In the next year he had...	3.81	−18.35	−1.67	0.644	−18.12	−1.65	0.665
BOOKING is essential for...	3.08	−56.67	−2.46	0.449	−56.93	−2.48	0.438

Question 2: Spearman Correlation [10 Marks]

Model	Score	Spearman ρ	p-value
Bigram	Total log-prob	0.1359	< 0.001
Bigram	Avg log-prob	0.0440	0.001
Bigram	SLOR	0.1177	< 0.001
Trigram	Total log-prob	**0.1387**	< 0.001
Trigram	Avg log-prob	0.0500	< 0.001
Trigram	SLOR	0.1227	< 0.001



All six correlations are statistically significant ($p < 0.01$), indicating that the language model scores are consistently associated with human acceptability judgements, although the correlations are relatively modest in magnitude. Among the evaluated metrics, the Trigram Total log-probability achieves the highest Spearman correlation ($\rho = 0.1387$), slightly outperforming the bigram model. This suggests that trigram models are better able to capture local syntactic context, making their probability estimates more sensitive to grammatical structure.

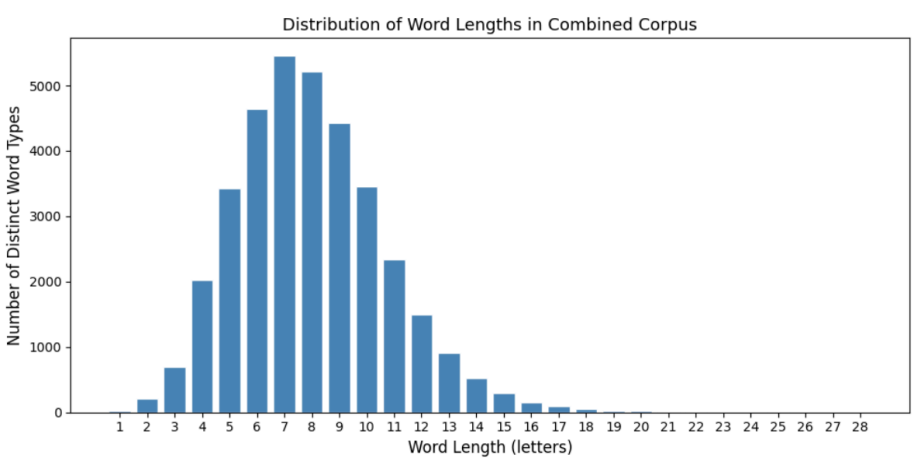
In contrast, SLOR does not outperform total log-probability in this dataset, unlike what has been reported in some previous studies. One possible reason is that the BNC sentences mainly consist of naturally occurring and generally well-formed text, with limited variation in syntactic acceptability, reducing the benefit of unigram normalisation. Average log-probability performs the weakest across both models, likely because normalising only by sentence length can unintentionally penalise sentences containing frequent short words, thereby conflating length effects with sentence quality.

Conclusion

N-gram language models have a weak but statistically significant relationship with human acceptability ratings. The trigram total log-probability is the best-performing combination from the n-gram family.

Question 3: Word Length, Frequency, and Information Content [20 marks]

3.1 Word Length Histogram



The histogram shows the number of distinct word types at each letter length. The distribution is right-skewed, peaking around 5–7 letters, with very few word types at extreme lengths. This is consistent with the general structure of the English lexicon.

3.2 Shortest Words

The shortest words in the dataset (length = 1) are:

Word	Frequency
a	16,089
i	4,836
s	210
c	176
b	147

The most common single-letter words are “a” and “i”, both of which are highly frequent in English. Their short length aligns with Zipf’s law, which suggests that frequently used words tend to become shorter to reduce communicative effort. Other single-letter entries such as “s”, “c”, and “b” are likely tokenisation artefacts, representing abbreviations, initials, or list markers rather than meaningful standalone words.

3.3 Pearson Correlation: Length vs Frequency

$Pearson(r) = -0.0641$

$p = 1.15 \times 10^{-33}$

The correlation is negative and highly significant, indicating that longer words tend to occur less frequently. Although the correlation is relatively small due to the highly skewed nature of word frequencies, the overall trend supports Zipf’s law of abbreviation: frequently used words tend to become shorter over time to reduce communicative effort.

3.4 Regression: Does IC Predict Word Length Better than Frequency?

Information content (IC) was defined as:

$$IC(w) = -log_2 P(w)$$

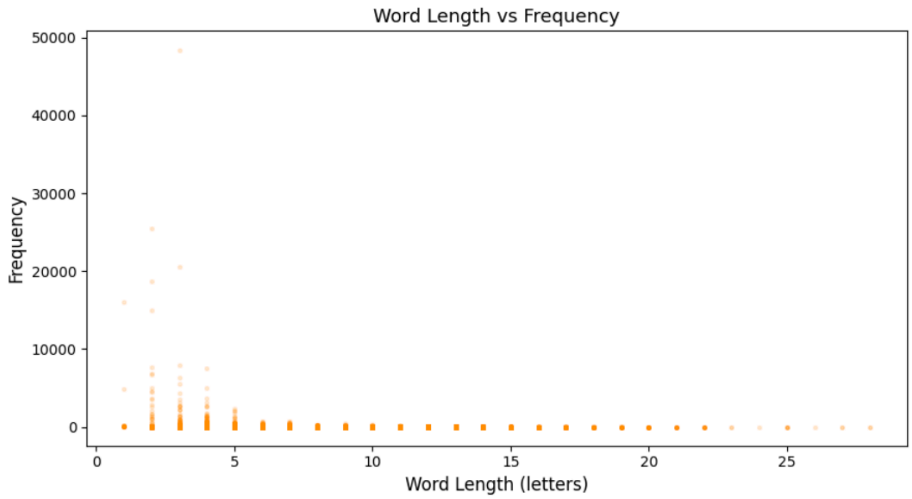
where P(w) = (count of w) / (total tokens). Higher IC indicates rarer, less predictable words.

Two linear regression models were fitted to predict word length:

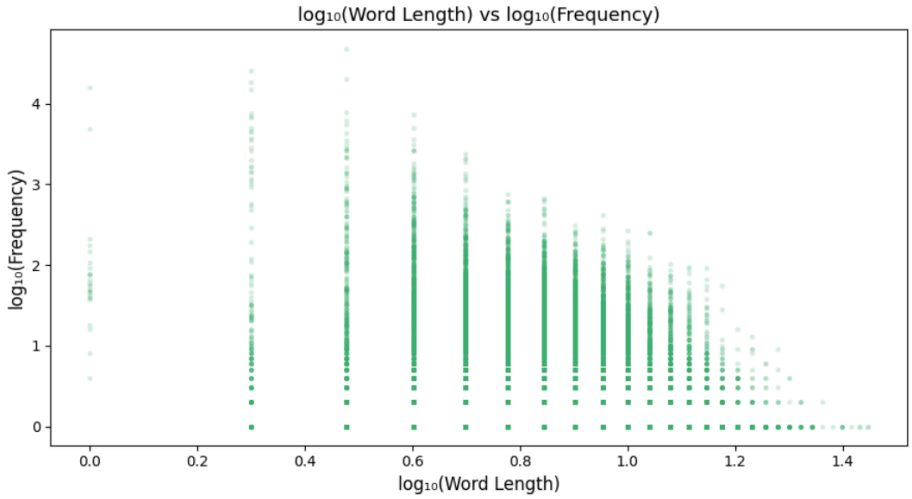
Predictor	R^2
Frequency	0.0041
Information Content (IC)	**0.0666**

Information content is a much stronger predictor of word length than raw frequency, with an R^2 value of 0.067 compared to 0.004 for frequency. This supports the hypothesis that information content is more closely related to word length and explains over 16 times more variance.

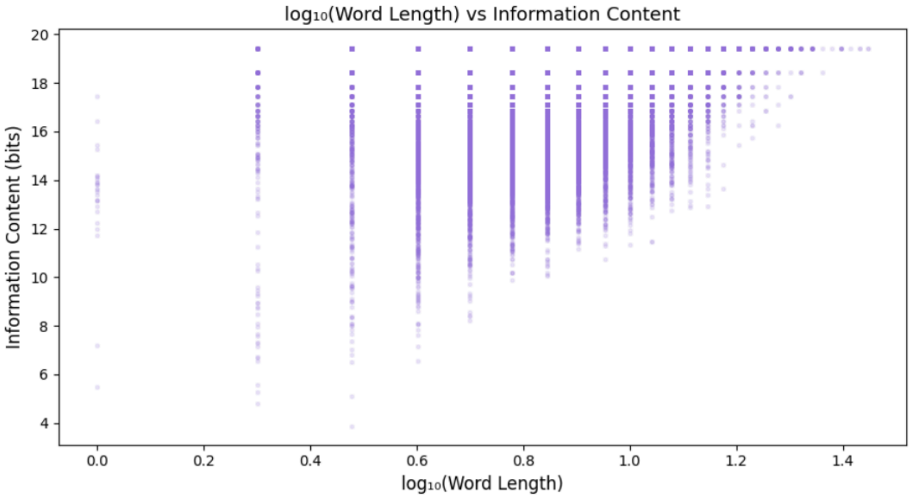
3.5 Plots



Word Length vs Frequency



log(Word Length) vs log(Frequency)



log(Word Length) vs Information Content

The log-log plot of word length versus frequency shows a clearer negative relationship than the raw plot, highlighting the underlying power-law pattern between the two variables. Similarly, the information content (IC) plot shows a positive trend, indicating that rarer words with higher IC values tend to be longer.

3.6 Are word lengths optimized for efficient communication?

The results of this analysis are consistent with the findings of Piantadosi, Tily & Gibson, which argue that word length is better predicted by information content (IC) than by raw frequency. Our regression results support this claim, with IC explaining substantially more variance in word length ($R^2 = 0.067$) compared to frequency alone ($R^2 = 0.004$). This suggests that communicative efficiency depends more on how predictable a word is in context than on how often it appears overall.

The results also show that shorter words tend to have lower information content and are therefore more predictable, while longer words usually carry more information. This pattern is visible in the scatter plot of log(length) versus IC. It reflects a broader principle of language efficiency, where the effort required to produce words is balanced against the amount of information they convey. In this sense, language behaves similarly to efficient coding systems such as Huffman coding in information theory.

Although the Pearson correlation between length and raw frequency is negative and statistically significant (−0.064), it remains relatively weak. This further supports the idea that frequency alone is not a sufficient measure of communicative load, whereas information content provides a more accurate explanation of variation in word length.

Extra Credit Questions [40 marks]

Model

GPT-2 (small, 117M parameters) was used, loaded from HuggingFace via `gpt2`. The model was run in evaluation mode on a GPU (T4). Sentence log-probabilities were extracted using the causal language modelling loss. Unigram log-probabilities were approximated using the corpus frequency counts from Question 3.

Spearman Correlation Results

Model	Score	Spearman ρ	p-value
Bigram	Total log-prob	0.1359	< 0.001
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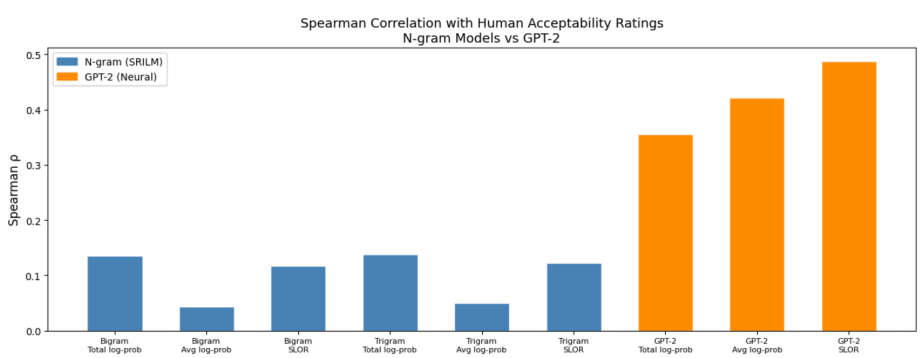
Trigram	Avg log-prob	0.0500	< 0.001
Trigram	SLOR	0.1227	< 0.001
GPT-2	**Total log-prob**	**0.3554**	**< 0.001**
GPT-2	**Avg log-prob**	**0.4213**	**< 0.001**
GPT-2	**SLOR**	**0.4876**	**< 0.001**



Discussion

GPT-2 performs substantially better than both n-gram models across all scoring metrics. The highest correlation is achieved by GPT-2 SLOR ($\rho = 0.4876$), which is almost four times higher than the best n-gram result obtained using trigram total log-probability ($\rho = 0.1387$). This indicates that GPT-2's probability estimates align much more closely with human acceptability judgements.

One major reason for this improvement is GPT-2's ability to use long-range context through self-attention, whereas bigram and trigram models can only consider one or two preceding words. Since sentence acceptability often depends on broader syntactic and semantic relationships, GPT-2 is better able to capture these patterns. In addition, GPT-2 uses byte-pair encoding (BPE), which allows it to handle rare or unseen words more effectively, while n-gram models assign extremely low probabilities to out-of-vocabulary words. GPT-2 was also trained on a very large and diverse corpus, enabling it to learn richer linguistic representations related to fluency and grammaticality.

Among the GPT-2 metrics, SLOR achieves the best performance, slightly outperforming average log-probability. This suggests that normalising for unigram frequency remains useful even for advanced neural language models. Overall, the results support the main hypothesis of the assignment: language model probability scores are correlated with human acceptability ratings, and stronger language models produce substantially higher correlations.